

Lecture 7

numerical integration,
rectangle, midpoint, trapezoidal, and Simpson's rules,
composite rules, interpolatory quadrature,
Clenshaw-Curtis and Gauss-Legendre quadratures

numerical integration

Example: arc length computation

Let us compute the length of a sinusoidal curve $y = \sin(x)$ spanning the interval $0 \leq x \leq 0.8$. According to the definition of the arc-length:

$$\begin{aligned} s &= \int_0^{0.8} \sqrt{1 + (y')^2} \, dx = \int_0^{0.8} \sqrt{1 + \cos^2(x)} \, dx = \sqrt{2} E \left(0.8 \left| \frac{1}{2} \right. \right) \\ &= 1.0759354360688678695640965157393553963837010239224760034752... \end{aligned}$$

This integral cannot be computed in terms of simple functions. It is written as another integral $E(\phi|k)$, called *incomplete elliptic integral of the second kind*, and then evaluated numerically (very precisely) using the iterative process called *arithmetic-geometric mean*.

In this lecture we study general methods of *numerical integration* that can be applied to evaluate this and other integrals.

Definition: Riemann integral

Let $[a, b]$ be an interval. A *partition* P_n of $[a, b]$ is a finite number of distinct points x_k such that $a = x_0 < x_1 < \dots < x_n = b$. Let T_n be a set of *intermediate points* t_k such that $x_{k-1} \leq t_k \leq x_k$. The lengths of the intervals are denoted as $h_k = x_k - x_{k-1}$, and the *mesh width* is defined as $m(P_n) = \max_{1 \leq k \leq n} h_k$. The *Riemann sum* R for a function $f \in C[a, b]$ is defined as:

$$R(f, P_n, T_n) = \sum_{k=1}^n h_k f(t_k)$$

Given the sequences $P_1, P_2, \dots$ and $T_1, T_2, \dots$ such that $\lim_{n \rightarrow \infty} m(P_n) = 0$, the function $f(x)$ is called *Riemann integrable* over $[a, b]$ if there exists the limit

$$\lim_{n \rightarrow \infty} R(f, P_n, T_n) = \mathcal{I},$$

which is called the *Riemann integral* of f over $[a, b]$:

$$\mathcal{I} = \int_a^b f(x) dx$$

Idea of numerical integration

All general numerical integration methods approximate the integral by a finite sum of the form:

$$\int_a^b f(x) dx \approx \sum_{k=1}^n w_k f(x_k)$$

The difference between the methods is in the choice of the points x_k (where the function $f(x)$ is evaluated) and the weights w_k .

Splitting the interval of integration

Let there be some partition P_n of the interval $[a, b]$. We shall use the following property of integrals:

$$\mathcal{I} = \int_a^b f(x) dx = \sum_{k=1}^n \int_{x_{k-1}}^{x_k} f(x) dx = \sum_{k=1}^n \mathcal{I}_k$$

First, we approximate each $\mathcal{I}_k$ (integration rules). Then, we see how the numerical errors add-up and what happens to the approximation of the total integral $\mathcal{I}$ when we refine the partition (composite rules).

rectangle rule

Left and right rectangle rules

$$\int_{x_{k-1}}^{x_k} f(x) dx \approx (x_k - x_{k-1})f(x_{k-1}) = h_k f(x_{k-1}) \quad (\text{left rectangle rule})$$

$$\int_{x_{k-1}}^{x_k} f(x) dx \approx (x_k - x_{k-1})f(x_k) = h_k f(x_k) \quad (\text{right rectangle rule})$$

How large is the error of these approximations?

Theorem: Error of the rectangle rule

Let $f \in C^1[x_{k-1}, x_k]$, $h_k = x_k - x_{k-1}$, and $m_1 = \max_{\xi \in [x_{k-1}, x_k]} |f'(\xi)|$. Then,

$$\left| \int_{x_{k-1}}^{x_k} f(x) dx - h_k f(x_{k-1}) \right| \leq \frac{1}{2} m_1 h_k^2$$
$$\left| \int_{x_{k-1}}^{x_k} f(x) dx - h_k f(x_k) \right| \leq \frac{1}{2} m_1 h_k^2$$

Proof

By the Taylor Theorem, there exists $\zeta(x) \in [x_{k-1}, x]$ such that $f(x) = f(x_{k-1}) + f'(\zeta(x))(x - x_{k-1})$. Hence,

$$\begin{aligned}\int_{x_{k-1}}^{x_k} f(x) dx &= \int_{x_{k-1}}^{x_k} [f(x_{k-1}) + f'(\zeta(x))(x - x_{k-1})] dx = \\ &= \int_{x_{k-1}}^{x_k} f(x_{k-1}) dx + \int_{x_{k-1}}^{x_k} f'(\zeta(x))(x - x_{k-1}) dx\end{aligned}$$

By the Mean-Value Theorem for Integration, since $x - x_{k-1} \geq 0$ for all $x \in [x_{k-1}, x_k]$, there exists $\eta \in [x_{k-1}, x_k]$ such that

$$\begin{aligned}\int_{x_{k-1}}^{x_k} f'(\zeta(x))(x - x_{k-1}) dx &= f'(\zeta(\eta)) \int_{x_{k-1}}^{x_k} (x - x_{k-1}) dx \\ &= f'(\xi) \int_{x_{k-1}}^{x_k} (x - x_{k-1}) dx\end{aligned}$$

Proof

Thus,

$$\begin{aligned}\int_{x_{k-1}}^{x_k} f(x) dx &= f(x_{k-1}) \int_{x_{k-1}}^{x_k} dx + f'(\xi) \int_{x_{k-1}}^{x_k} (x - x_{k-1}) dx \\&= h_k f(x_{k-1}) + f'(\xi) \frac{1}{2} u^2 \Big|_0^{h_k} \\&= h_k f(x_{k-1}) + f'(\xi) \frac{1}{2} h_k^2\end{aligned}$$

Therefore,

$$\left| \int_{x_{k-1}}^{x_k} f(x) dx - h_k f(x_{k-1}) \right| = \frac{1}{2} |f'(\xi)| h_k^2 \leq \frac{1}{2} \max_{\xi \in [x_{k-1}, x_k]} |f'(\xi)| h_k^2$$

Proof for the right rectangle rule is similar. □

midpoint rule

Midpoint rule

$$\int_{x_{k-1}}^{x_k} f(x) dx \approx h_k f(x_m),$$
$$x_m = \frac{x_{k-1} + x_k}{2}, \quad (\text{midpoint})$$

Theorem: Error of the midpoint rule

Let $f \in C^2[x_{k-1}, x_k]$, $h_k = x_k - x_{k-1}$, $x_m = (x_{k-1} + x_k)/2$, and $m_2 = \max_{\xi \in [x_{k-1}, x_k]} |f''(\xi)|$. Then,

$$\left| \int_{x_{k-1}}^{x_k} f(x) dx - h_k f(x_m) \right| \leq \frac{1}{24} m_2 h_k^3$$

Proof

By the Taylor Theorem, there exists $\zeta(x)$ between x and x_m such that

$$f(x) = f(x_m) + f'(x_m)(x - x_m) + \frac{f''(\zeta(x))}{2}(x - x_m)^2$$

Hence,

$$\begin{aligned} \int_{x_{k-1}}^{x_k} f(x) dx &= \int_{x_{k-1}}^{x_k} \left[f(x_m) + f'(x_m)(x - x_m) + \frac{f''(\zeta(x))}{2}(x - x_m)^2 \right] dx = \\ &= h_k f(x_m) + \int_{x_{k-1}}^{x_k} f'(x_m)(x - x_m) dx + \frac{1}{2} \int_{x_{k-1}}^{x_k} f''(\zeta(x))(x - x_m)^2 dx \end{aligned}$$

We compute:

$$\begin{aligned} \int_{x_{k-1}}^{x_k} f'(x_m)(x - x_m) dx &= f'(x_m) \int_{x_{k-1}}^{x_k} (x - x_m) dx \\ &= f'(x_m) \frac{1}{2} u^2 \Big|_{-h_k/2}^{h_k/2} = \frac{f'(x_m)}{2} \left[\left(\frac{h_k}{2} \right)^2 - \left(-\frac{h_k}{2} \right)^2 \right] \\ &= \frac{f'(x_m)}{2} \left[\left(\frac{h_k}{2} \right)^2 - \left(\frac{h_k}{2} \right)^2 \right] = 0 \end{aligned}$$

Proof

By the Mean-Value Theorem for Integration, since $(x - x_m)^2 \geq 0$ for all $x \in [x_{k-1}, x_k]$, there exists $\eta \in [x_{k-1}, x_k]$ such that

$$\begin{aligned} \frac{1}{2} \int_{x_{k-1}}^{x_k} f''(\zeta(x))(x - x_m)^2 dx &= f''(\zeta(\eta)) \left. \frac{1}{6} u^3 \right|_{-h_k/2}^{h_k/2} \\ &= \frac{f''(\xi)}{6} \left[\left(\frac{h_k}{2} \right)^3 - \left(-\frac{h_k}{2} \right)^3 \right] = \frac{f''(\xi)}{6} \left[\frac{h_k^3}{8} + \frac{h_k^3}{8} \right] \\ &= \frac{f''(\xi)}{24} h_k^3 \end{aligned}$$

Proof

Thus,

$$\int_{x_{k-1}}^{x_k} f(x) dx = h_k f(x_m) + \frac{f''(\xi)}{24} h_k^3$$

Therefore,

$$\left| \int_{x_{k-1}}^{x_k} f(x) dx - h_k f(x_m) \right| = \frac{|f''(\xi)|}{24} h_k^3 \leq \frac{1}{24} \max_{\xi \in [x_{k-1}, x_k]} |f''(\xi)| h_k^3$$



trapezoidal rule

Trapezoidal rule

$$\int_{x_{k-1}}^{x_k} f(x) dx \approx \frac{h_k}{2} f(x_{k-1}) + \frac{h_k}{2} f(x_k)$$

Theorem: Error of the trapezoidal rule

Let $f \in C^2[x_{k-1}, x_k]$, $h_k = x_k - x_{k-1}$, $x_m = (x_{k-1} + x_k)/2$, and $m_2 = \max_{\xi \in [x_{k-1}, x_k]} |f''(\xi)|$. Then,

$$\left| \int_{x_{k-1}}^{x_k} f(x) dx - \frac{h_k}{2}(f(x_k) + f(x_{k-1})) \right| \leq \frac{1}{12} m_2 h_k^3$$

Proof

First, we show that the trapezoidal rule is obtained by integrating a linear approximation (interpolation) of $f(x)$ over $[x_{k-1}, x_k]$. By definition, the linear interpolation is:

$$L_1(x) = \frac{x_k - x}{x_k - x_{k-1}} f(x_{k-1}) + \frac{x - x_{k-1}}{x_k - x_{k-1}} f(x_k)$$

Computing the integral of $L(x)$:

$$\begin{aligned} \int_{x_{k-1}}^{x_k} L_1(x) dx &= \frac{f(x_{k-1})}{h_k} \int_{x_{k-1}}^{x_k} (x_k - x) dx + \frac{f(x_k)}{h_k} \int_{x_{k-1}}^{x_k} (x - x_{k-1}) dx \\ &= \frac{f(x_{k-1})}{h_k} \left[-\frac{1}{2} u^2 \right]_{h_k}^0 + \frac{f(x_k)}{h_k} \left[\frac{1}{2} u^2 \right]_0^{h_k} = \frac{h_k}{2} f(x_{k-1}) + \frac{h_k}{2} f(x_k) \end{aligned}$$

Hence, the error of the trapezoidal method is:

$$\begin{aligned} \left| \int_{x_{k-1}}^{x_k} f(x) dx - \frac{h_k}{2} (f(x_k) + f(x_{k-1})) \right| &= \left| \int_{x_{k-1}}^{x_k} f(x) dx - \int_{x_{k-1}}^{x_k} L_1(x) dx \right| \\ &= \left| \int_{x_{k-1}}^{x_k} [f(x) - L_1(x)] dx \right| \end{aligned}$$

Recall from lecture about Interpolation:

Theorem: Error of linear approximation

Let $x_0, x_1 \in [a, b]$, $x_0 < x_1$, and $f(x) \in C^2[a, b]$. If $L_1(x)$ is the linear polynomial, such that $L_1(x_0) = f(x_0)$ and $L_1(x_1) = f(x_1)$, then for each $x \in [a, b]$ there exists $\xi \in (a, b)$ such that:

$$f(x) - L_1(x) = \frac{f''(\xi)}{2}(x - x_0)(x - x_1).$$

Hence, we can write:

$$\int_{x_{k-1}}^{x_k} [f(x) - L_1(x)] dx = \int_{x_{k-1}}^{x_k} \frac{f''(\xi)}{2}(x - x_{k-1})(x - x_k) dx$$

where $\xi \in [x_k, x_{k-1}]$

Proof

$$\begin{aligned} \int_{x_{k-1}}^{x_k} f(x) dx - \frac{h_k}{2}(f(x_k) + f(x_{k-1})) &= \frac{f''(\xi)}{2} \int_{x_{k-1}}^{x_k} (x - x_{k-1})(x - x_k) dx \\ &= \frac{f''(\xi)}{2} \int_{x_{k-1}}^{x_k} (x - x_k + x_k - x_{k-1})(x - x_k) dx \\ &= \frac{f''(\xi)}{2} \int_{x_{k-1}}^{x_k} (x - x_k + h_k)(x - x_k) dx = \frac{f''(\xi)}{2} \int_{x_{k-1}}^{x_k} [(x - x_k)^2 + h_k(x - x_k)] dx \\ &= \frac{f''(\xi)}{2} \left[\frac{1}{3} u^3 + \frac{h_k}{2} u^2 \right]_{h_k}^0 = \frac{f''(\xi)}{2} \left(\frac{1}{3} h_k^3 - \frac{h_k}{2} h_k^2 \right) = -\frac{f''(\xi)}{12} h_k^3 \end{aligned}$$

Hence,

$$\left| \int_{x_{k-1}}^{x_k} f(x) dx - \frac{h_k}{2}(f(x_k) + f(x_{k-1})) \right| = \frac{|f''(\xi)|}{12} h_k^3 \leq \frac{1}{12} \max_{\xi \in [x_{k-1}, x_k]} |f''(\xi)| h_k^3$$



Simpson's rule

Using interpolation for numerical integration

Trapezoidal rule is based on the linear interpolation (approximation of f)
 $f(x) \approx L_1(x)$:

$$\int_{x_{k-1}}^{x_k} f(x) dx \approx \int_{x_{k-1}}^{x_k} L_1(x) dx = \frac{h_k}{2} f(x_{k-1}) + \frac{h_k}{2} f(x_k)$$

By the same logic we can use a higher-order interpolation to derive a hopefully higher-order integration rule. For example, with $f(x) \approx L_2(x)$ we get:

$$\int_{x_{k-1}}^{x_k} f(x) dx \approx \int_{x_{k-1}}^{x_k} L_2(x) dx$$

Higher-order interpolations require more function evaluation points. Quadratic interpolation needs three points. The easiest choice is x_{k-1} , x_m , and x_k .

Recall from lecture about Interpolation:

Definition: Lagrange polynomial

Given $n + 1$ distinct data points x_i and $n + 1$ function values $f(x_i)$, $i = 0, \dots, n$, the *Lagrange interpolating polynomial* is the n -th degree polynomial $L_n(x)$ that passes through all function values and has the form:

$$L_n(x) = \sum_{j=0}^n f(x_j) L_{j,n}(x),$$

where the *Lagrange basis functions* are given by:

$$L_{k,n}(x) = \frac{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)}$$

Now we use it to construct $L_2(x)$

Derivation of Simpson's rule

With $x_0 = x_{k-1}$, $x_1 = x_m$, and $x_2 = x_k$, we get:

$$L_2(x) = \sum_{j=0}^2 f(x_j)L_{j,2}(x) = f(x_{k-1})L_{k-1,2}(x) + f(x_m)L_{m,2}(x) + f(x_k)L_{k,2}(x),$$

$$L_{k-1,2}(x) = \frac{(x - x_m)(x - x_k)}{(x_{k-1} - x_m)(x_{k-1} - x_k)} = \frac{2}{h_k^2}(x - x_m)(x - x_k)$$

$$L_{m,2}(x) = \frac{(x - x_{k-1})(x - x_k)}{(x_m - x_{k-1})(x_m - x_k)} = -\frac{4}{h_k^2}(x - x_{k-1})(x - x_k)$$

$$L_{k,2}(x) = \frac{(x - x_{k-1})(x - x_m)}{(x_k - x_{k-1})(x_k - x_m)} = \frac{2}{h_k^2}(x - x_{k-1})(x - x_m)$$

Derivation of Simpson's rule

$$\begin{aligned}\int_{x_{k-1}}^{x_k} L_2(x) dx &= f(x_{k-1}) \int_{x_{k-1}}^{x_k} L_{k-1,2}(x) dx \\ &\quad + f(x_m) \int_{x_{k-1}}^{x_k} L_{m,2}(x) dx \\ &\quad + f(x_k) \int_{x_{k-1}}^{x_k} L_{k,2}(x) dx\end{aligned}$$

$$\begin{aligned}\int_{x_{k-1}}^{x_k} L_{k-1,2}(x) dx &= \frac{2}{h_k^2} \int_{x_{k-1}}^{x_k} (x - x_m)(x - x_k) dx \\ &= \frac{2}{h_k^2} \int_{x_{k-1}}^{x_k} \left(x - x_k + \frac{h_k}{2}\right)(x - x_k) dx = \frac{2}{h_k^2} \int_{x_{k-1}}^{x_k} \left[(x - x_k)^2 + \frac{h_k}{2}(x - x_k)\right] dx \\ &= \frac{2}{h_k^2} \left[\frac{1}{3}u^3 + \frac{h_k}{4}u^2\right]_{-h_k}^0 = \frac{2}{h_k^2} \left[-\frac{1}{3}(-h_k)^3 - \frac{h_k}{4}(-h_k)^2\right] = \frac{1}{6}h_k\end{aligned}$$

Derivation of Simpson's rule

$$\begin{aligned}\int_{x_{k-1}}^{x_k} L_{m,2}(x) dx &= -\frac{4}{h_k^2} \int_{x_{k-1}}^{x_k} (x - x_{k-1})(x - x_k) dx \\&= -\frac{4}{h_k^2} \int_{x_{k-1}}^{x_k} (x - x_k + h_k)(x - x_k) dx = -\frac{4}{h_k^2} \int_{x_{k-1}}^{x_k} [(x - x_k)^2 + h_k(x - x_k)] dx \\&= -\frac{4}{h_k^2} \left[\frac{1}{3} u^3 + \frac{h_k}{2} u^2 \right]_{-h_k}^0 = -\frac{4}{h_k^2} \left[\frac{1}{3} h_k^3 - \frac{1}{2} h_k^3 \right] = \frac{4}{6} h_k\end{aligned}$$

$$\int_{x_{k-1}}^{x_k} L_{k,2}(x) dx = \frac{1}{6} h_k$$

Simpson's rule

$$\int_{x_{k-1}}^{x_k} f(x) dx \approx \frac{h_k}{6} f(x_{k-1}) + \frac{4h_k}{6} f(x_m) + \frac{h_k}{6} f(x_k),$$
$$x_m = \frac{x_{k-1} + x_k}{2}$$

Theorem: Error of Simpson's rule

Let $f \in C^4[x_{k-1}, x_k]$, $h_k = x_k - x_{k-1}$, $x_m = (x_{k-1} + x_k)/2$, and $m_4 = \max_{\xi \in [x_{k-1}, x_k]} |f^{(4)}(\xi)|$. Then,

$$\left| \int_{x_{k-1}}^{x_k} f(x) dx - \frac{h_k}{6} (f(x_k) + 4f(x_m) + f(x_{k-1})) \right| \leq \frac{1}{2880} m_4 h_k^5$$

Proof. Compute the integral of the $L_2(x)$ interpolation error.

Questions

- Finish the derivation of the Simpson's rule on Slide 29 by computing the integral of $L_{k,2}(x)$

composite rules

Approximating integral by composition

Let there be some partition of the interval $[a, b]$ into sub-intervals. We used the following property of integrals:

$$\mathcal{I} = \int_a^b f(x) dx = \sum_{k=1}^n \int_{x_{k-1}}^{x_k} f(x) dx = \sum_{k=1}^n \mathcal{I}_k$$

- To approximate each $\mathcal{I}_k \approx I_k$, we have introduced common numerical integration rules (rectangle, midpoint, trapezoidal, Simpson's) and studied their errors, $|\mathcal{I}_k - I_k|$
- Now we study *composite rules*. The main question: What is the error $|\mathcal{I} - I|$ when $\mathcal{I}$ is approximated as:

$$\mathcal{I} \approx I = \sum_{k=1}^n I_k,$$

where I_k is the approximation of $\mathcal{I}_k$ by one of the common integration rules?

Theorem: Error of the composite rule

Let $f(x)$ be given for $x \in [a, b]$, and let there be a *uniform* partition $a = x_0 < x_1 < \dots < x_n = b$, with $x_k = x_0 + kh$, $k = 0, \dots, n$, $h = (b - a)/n$. Let I_k be the approximation of the integral over $[x_{k-1}, x_k]$, such that

$$\left| I_k - \int_{x_{k-1}}^{x_k} f(x) dx \right| \leq c_k h^{p+1}, \quad p \in \mathbb{N}.$$

Then,

$$\left| \sum_{k=1}^n I_k - \int_a^b f(x) dx \right| \leq c(b - a)h^p$$

where $c = \max_k c_k$.

$$\begin{aligned} \left| \sum_{k=1}^n I_k - \int_a^b f(x) dx \right| &= \left| \sum_{k=1}^n I_k - \sum_{k=1}^n \int_{x_{k-1}}^{x_k} f(x) dx \right| \\ &= \left| \sum_{k=1}^n \left(I_k - \int_{x_{k-1}}^{x_k} f(x) dx \right) \right| \leq \sum_{k=1}^n \left| I_k - \int_{x_{k-1}}^{x_k} f(x) dx \right| \\ &\leq \sum_{k=1}^n c_k h^{p+1} = h^{p+1} \sum_{k=1}^n c_k \leq h^{p+1} \max_k c_k \sum_{k=1}^n (1) = ch^{p+1} n \\ &= ch^{p+1} \frac{b-a}{h} = c(b-a)h^p \end{aligned}$$



Errors of common composite rules

integration rule	$ I_k - \mathcal{I}_k \leq$	$ \sum_{k=1}^n I_k - \mathcal{I} \leq$
left/right rectangle	$\frac{1}{2} m_{1,k} h^2$	$\frac{1}{2} M_1(b-a)h$
midpoint	$\frac{1}{24} m_{2,k} h^3$	$\frac{1}{24} M_2(b-a)h^2$
trapezoidal	$\frac{1}{12} m_{2,k} h^3$	$\frac{1}{12} M_2(b-a)h^2$
Simpson's	$\frac{1}{2880} m_{4,k} h^5$	$\frac{1}{2880} M_4(b-a)h^4$

Assuming uniform partition of $[a, b]$ with $h = (b - a)/n$ and using the notation:

$$m_{n,k} = \max_{x \in [x_{k-1}, x_k]} |f^{(n)}(x)|$$

$$M_n = \max_k m_{n,k} = \max_{x \in [a, b]} |f^{(n)}(x)|$$

Example: using error bounds to find good h

For the length of a sinusoidal curve $y = \sin(x)$ spanning the interval $0 \leq x \leq 0.8$ we have the "exact" result:

$$s = \int_0^{0.8} \sqrt{1 + \cos^2(x)} dx = \sqrt{2}E\left(0.8 \left| \frac{1}{2} \right.\right) \\ = 1.0759354360688678695640965157393553963837010239224760034752...$$

- Suppose we want to use the *composite left rectangle rule* to compute this integral numerically. You can write a program and run it, but first...
- Question 1: How small should h be, to have the approximation error of at most 0.01?
- Question 2: How many function evaluations are required for the approximation error to be at most 0.01?

Example: using error bounds to find good h

From the table of error bounds, for the composite left rectangle rule we find:

$$\left| \sum_{k=1}^n I_k - \mathcal{I} \right| \leq \frac{1}{2} M_1 (b-a)h, \quad M_1 = \max_{x \in [a,b]} |f'(x)|$$

In this case $b - a = 0.8 - 0 = 0.8$, and:

$$\begin{aligned} M_1 &= \max_{x \in [0,0.8]} \left| \left(\sqrt{1 + \cos^2(x)} \right)' \right| = \max_{x \in [0,0.8]} \left| \frac{-2 \cos(x) \sin(x)}{2 \sqrt{1 + \cos^2(x)}} \right| \\ &= \max_{x \in [0,0.8]} \left| \frac{\sin(2x)}{2 \sqrt{1 + \cos^2(x)}} \right| \leq \frac{\max_{x \in [0,0.8]} |\sin(2x)|}{2 \min_{x \in [0,0.8]} \sqrt{1 + \cos^2(x)}} \leq \frac{1}{2} \end{aligned}$$

Example: using error bounds to find good h

Hence, we get the error below 0.01, i.e.,

$$\left| \sum_{k=1}^n I_k - \mathcal{I} \right| \leq 0.01,$$

if we choose h so that the upper bound on the error is below 0.01:

$$\begin{aligned} \frac{1}{2} M_1(b-a)h &\leq 0.01 \\ \frac{1}{2} \frac{1}{2} (0.8 - 0)h &= \frac{0.8}{4} h = 0.2h \leq 0.01 \\ h &\leq \frac{0.01}{0.2} = 0.05 \end{aligned}$$

Answer: the composite left rectangle rule will require $h \leq 0.05$ for the (upper bound of) error to be at most 0.01.

Example: using error bounds to estimate n

How many function evaluations will the approximation with $h \leq 0.05$ require?

Composite left rectangle rule requires n function values, specifically, $f(x_0), f(x_1), \dots, f(x_{n-1})$. From the condition on h , we obtain the condition on n :

$$\begin{aligned}h &= \frac{b-a}{n} \leq 0.05 \\ \frac{0.8}{n} &\leq 0.05 \\ \frac{1}{n} &\leq \frac{0.05}{0.8} \\ n &\geq \frac{0.8}{0.05} = \text{round this up to the first integer} = 16\end{aligned}$$

Answer: the composite left rectangle rule will require *at least* 16 function evaluations for the (upper bound of) error to be at most 0.01.

interpolatory quadrature

Interpolation instead of composition?

- We saw that interpolating a function $f(x)$ on $[x_{k-1}, x_k]$ produced higher-order integration rules, like trapezoidal and Simpson's, depending on how many nodes were used
- Now that we have the partition of $[a, b]$: $a = x_0 < x_1 < \dots < x_n = b$ with $n + 1$ nodes, we can also make a new global quadrature rule by interpolating $f(x)$ over $[a, b]$ using all the nodes of the partition
- Question 1: What kind of functions f are integrated *exactly* with this new interpolating quadrature?
- Question 2: What is the error of this quadrature for general functions f ?

Interpolation on the nodes of partition

- We have the partition of $[a, b]$: $a = x_0 < x_1 < \dots < x_n = b$, i.e., $n + 1$ nodes x_k , $k = 0, \dots, n$.
- We recall that to determine all $n + 1$ coefficients of the n -th degree polynomial

$$L_n(x) = c_0 + c_1x + \dots + c_nx^n$$

we need at least $n + 1$ points

- Hence, it makes sense to interpolate f with L_n :

$$f(x) \approx L_n(x), \quad x \in [a, b]$$

and to approximate the integral as:

$$\int_a^b f(x) dx \approx \int_a^b L_n(x) dx$$

Lagrange form of L_n

To compute the integral of the approximation L_n of f , we use the Lagrange form of the interpolating polynomial L_n :

$$L_n(x) = \sum_{k=0}^n f(x_k) L_{k,n}(x),$$

where the *Lagrange basis functions* are given by:

$$L_{k,n}(x) = \frac{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)}$$

Interpolatory quadrature

The approximation of the integral becomes:

$$\begin{aligned}\int_a^b f(x) dx &\approx \int_a^b L_n(x) dx = \int_a^b \sum_{k=0}^n f(x_k) L_{k,n}(x) dx \\ &= \sum_{k=0}^n f(x_k) \int_a^b L_{k,n}(x) dx\end{aligned}$$

Which has the form of a quadrature rule

$$\int_a^b f(x) dx \approx \sum_{k=0}^n w_k f(x_k),$$

with the weights:

$$w_k = \int_a^b L_{k,n}(x) dx$$

Computing these w_k requires some analytical effort and is offered in many software packages.

Theorem: Exactness of interpolatory quadrature

Let $f(x)$ be given for $x \in [a, b]$. Let there be a partition with the nodes $a = x_0 < \dots < x_n = b$, and the interpolating polynomial $L_n(x)$ of $f(x)$ on $[a, b]$. Then, the interpolatory quadrature:

$$\sum_{k=0}^n w_k f(x_k), \quad w_k = \int_a^b L_{k,n}(x) dx,$$

where $L_{k,n}(x)$ are the Lagrange basis functions, equals the exact value of the integral

$$\mathcal{I} = \int_a^b f(x) dx$$

if $f(x)$ is a polynomial of degree at most n .

Theorem: Error of interpolatory quadrature

Let $f(x)$ be given for $x \in [a, b]$. Let there be a partition with the nodes $a = x_0 < \dots < x_n = b$, and the interpolating polynomial $L_n(x)$ of $f(x)$ on $[a, b]$. If n is even and $f \in C^{n+2}[a, b]$, then there is $\xi \in [a, b]$ such that

$$\left| \int_a^b f(x) dx - \int_a^b L_n(x) dx \right| \leq C_n \left(\frac{b-a}{n} \right)^{n+3} |f^{(n+2)}(\xi)|,$$
$$C_n = \frac{1}{(n+2)!} \int_0^n t^2(t-1) \cdots (t-n) dt.$$

If n is odd and $f \in C^{n+1}[a, b]$, then there is $\xi \in [a, b]$ such that

$$\left| \int_a^b f(x) dx - \int_a^b L_n(x) dx \right| \leq D_n \left(\frac{b-a}{n} \right)^{n+2} |f^{(n+1)}(\xi)|,$$
$$D_n = \frac{1}{(n+1)!} \int_0^n t(t-1) \cdots (t-n) dt.$$

Nodes from uniform partition: Newton-Cotes quadrature

With the uniformly distributed nodes $x_k = a + kh$, $h = (b - a)/n$, the interpolatory quadrature

$$\int_a^b f(x) dx \approx \sum_{k=0}^n w_k f(x_k), \quad w_k = \int_a^b L_{k,n}(x) dx$$

is called the *Newton-Cotes* quadrature.

- Trapezoidal rule on $[a, b]$ is a Newton-Cotes quadrature with $n = 1$; step $h = b - a$; nodes $x_0 = a$, $x_1 = a + h = b$; weights $w_0 = w_1 = h/2$
- Simpson's rule on $[a, b]$ is a Newton-Cotes quadrature with $n = 2$; step $h = (b - a)/2$; nodes $x_0 = a$, $x_1 = a + h$, $x_2 = a + 2h = b$; weights $w_0 = w_2 = h/3$, $w_1 = 4h/3$
- Higher-order, $n > 2$, Newton-Cotes have negative weights and are rarely used
- Newton-Cotes quadrature of degree n will obey the Exactness Theorem, but often gives very large error for polynomials of degrees higher than n . Related to Runge phenomenon.

Clenshaw-Curtis and Gauss-Legendre quadratures

Tuning the partition

- Given the uniformly distributed nodes on $[a, b]$: $x_k = a + kh$, $k = 0, \dots, n$, $h = (b - a)/n$, the Newton-Cotes quadrature with $n > 2$ may result in large errors for general functions f .
- What if we could choose the location of the nodes on $[a, b]$?
- We know that the interpolation error $|f(x) - L_n(x)|$ depends on the choice of nodes, see Runge phenomenon
- Hence, interpolatory quadrature will also depend on the choice of nodes

Chebyshev nodes: Clenshaw-Curtis quadrature

The interpolation error $|f(x) - L_n(x)|$ is uniformly minimized over $[a, b]$ by choosing the *Chebyshev nodes* on $[a, b]$:

$$x_k = \frac{a+b}{2} + \frac{b-a}{2} \cos\left(\frac{2k+1}{2(n+1)}\pi\right), \quad k = 0, \dots, n$$

- The interpolatory quadrature obtained with this choice of nodes is called the *Clenshaw-Curtis* quadrature
- The weights w_k are always positive
- Quadrature is exact for polynomials of degree at most n (because it is an interpolatory quadrature, see Exactness Theorem)
- Error remains small for polynomials of degrees up to $2n$

Legendre nodes: Gauss-Legendre quadrature

- Obtained (Gauss, 1814) by tuning both the nodes x_k and the weights w_k to maximize the degree of the polynomial for which the quadrature is exact with n nodes
- This quadrature uses just n nodes on $[a, b]$ but is exact for polynomials of degree at most $2n - 1$
- This is the maximum *exactness* one can achieve with n nodes
- Turns out that the nodes are the roots of the degree n Legendre polynomial $P_n(x)$, i.e., all x_k for which $P_n(x_k) = 0$.
- The weights w_k are always positive
- Gauss-Legendre quadrature (nodes and weights) are usually defined for integrals over $[-1, 1]$, but this can be transformed to the general interval $[a, b]$ by variable change in your integral
- Nodes and weights of the Gauss-Legendre quadrature on $[-1, 1]$ are generated by software packages, e.g., `scipy.special.p_roots()`

Summary: composite quadratures

- Quadrature rule with $n + 1$ nodes on $[x_{k-1}, x_k]$:

$$\int_{x_{k-1}}^{x_k} f(x) dx \approx \sum_{j=0}^n w_j f(x_j)$$

with nodes x_j and weights w_j .

- Composite quadratures for integrals over $[a, b]$ are obtained by applying common quadrature rules on each sub-interval of the partition $a = x_0 < x_1 < \dots < x_n = b$, and adding up the results
- Error of the composite quadrature is one order lower, $\mathcal{O}(h^{p-1})$, than the error of the quadrature rule applied, $\mathcal{O}(h^p)$
- Composite quadrature is used when the nodes/partition are given or fixed
- You should be able to estimate the step size h and the number of partitions n that will guarantee the error to be smaller than a given ϵ (using the table of errors)

Summary: interpolatory quadratures

- Interpolatory quadratures on $n + 1$ nodes:

$$\int_a^b f(x) dx \approx \int_a^b L_n(x) dx = \sum_{k=0}^n f(x_k) \int_a^b L_{k,n}(x) dx$$

where $L_n(x)$ is the interpolating polynomial of $f(x)$.

- The weights w_k of the interpolating quadrature are the integrals of the Lagrange basis functions over $[a, b]$
- Any interpolatory quadrature with $n + 1$ nodes is exact for polynomials of degree at most n
- Newton-Cotes quadrature (uniform nodes) may produce large errors for polynomials with degrees higher than n
- Clenshaw-Curtis quadrature (Chebyshev nodes) has small error for polynomials of degree up to $2n$
- Gauss-Legendre quadrature (Legendre nodes) is exact for polynomials of degree up to $2n - 1$

Questions

- Suppose that you want to use the composite midpoint rule for integrating some function $f(x)$ over $[a, b]$ and you want the error to be smaller than ϵ . How to determine the required step h ? You may use the table on Slide 6